

Classical & Quantum Cramér–Rao Bounds

Question & Answer Study Guide

From Statistics to Quantum Metrology

“Quantum estimation generalizes classical estimation by optimizing over all possible measurements.”

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1 Classical Cramér–Rao Bound

Q1 — What is the Classical Cramér–Rao Bound?

Question 1

Define the classical Cramér–Rao Bound (CRB) and state its mathematical form. What does it fundamentally tell us about estimation?

Answer

The **Cramér–Rao Bound** (CRB) is a foundational result in mathematical statistics that establishes a hard lower limit on how precisely we can estimate an unknown parameter θ from observed data.

Setup: Suppose data x is drawn from a parametric distribution $p(x \mid \theta)$, and $\hat{\theta}(x)$ is any *unbiased* estimator, meaning

$$\mathbb{E}[\hat{\theta}] = \theta.$$

The bound states:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)} \quad (1)$$

where $I(\theta)$ is the **Fisher Information**:

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln p(x \mid \theta) \right)^2 \right]. \quad (2)$$

Interpretation: No matter how clever your estimator, its variance can never fall below $1/I(\theta)$. Equality in (1) is achieved if and only if the estimator is *efficient* (e.g. the maximum-likelihood estimator for exponential family distributions).

Intuition for the Fisher score: The quantity $\frac{\partial}{\partial \theta} \ln p$ is called the *score*. It measures how rapidly the log-likelihood changes when θ changes. A large score variance $\Rightarrow$ large $I(\theta) \Rightarrow$ tight CRB $\Rightarrow$ easy to estimate.

Key Insight

Think of Fisher Information as a “signal-to-noise” gauge for the parameter. High $I(\theta)$ means the data are very sensitive to the parameter’s value, so we can pin it down precisely. Low $I(\theta)$ means the data look almost the same regardless of θ , making estimation difficult.

Q2 — What is an *efficient* estimator?

Question 2

What conditions must be satisfied for an estimator to be called *efficient*, and why is efficiency an important concept in statistics and signal processing?

Answer

An estimator $\hat{\theta}$ is called **efficient** when it:

1. Is *unbiased*: $\mathbb{E}[\hat{\theta}] = \theta$ for all θ .
2. *Achieves equality* in the CRB:

$$\text{Var}(\hat{\theta}) = \frac{1}{I(\theta)}.$$

Characterisation: Equality in the CRB is achieved if and only if the score satisfies a linear relationship:

$$\frac{\partial}{\partial \theta} \ln p(x | \theta) = I(\theta)(\hat{\theta}(x) - \theta).$$

This is satisfied for all members of the *exponential family* of distributions (Gaussian, Poisson, Bernoulli, Gamma, ...).

Why it matters:

- An efficient estimator wastes *zero* information contained in the data.
- It gives the tightest confidence intervals possible.
- In radar, communications, and ML, efficient estimators directly translate to the minimum possible mean-squared error.

Example: For $x_1, \dots, x_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ^2 , the sample mean $\bar{x} = \frac{1}{n} \sum x_i$ is efficient with $\text{Var}(\bar{x}) = \sigma^2/n = 1/I(\mu)$.

Q3 — How does Fisher Information scale with sample size?

Question 3

If we have n independent, identically distributed (i.i.d.) observations, how does the total Fisher information relate to the per-sample Fisher information? What are the consequences for the CRB?

Answer

For n i.i.d. observations $x_1, \dots, x_n$ each drawn from $p(x \mid \theta)$:

$$I_n(\theta) = n I_1(\theta) \quad (3)$$

This follows directly from the *additivity* of Fisher information: the joint log-likelihood decomposes as a sum, and differentiation distributes term-by-term.

Consequence for the CRB:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n I_1(\theta)}$$

This shows:

- The minimum achievable standard deviation scales as $1/\sqrt{n}$ (the celebrated “ $\sqrt{n}$ law” or “shot-noise scaling”).
- Doubling sample size reduces the best possible standard deviation by a factor of $\sqrt{2}$.
- This is the *classical* scaling limit — quantum resources can potentially beat it (see Section 2).

Graphical Summary:

n	Fisher Info I_n	Best Std Dev
1	I_1	$1/\sqrt{I_1}$
4	$4I_1$	$1/(2\sqrt{I_1})$
100	$100I_1$	$1/(10\sqrt{I_1})$

Shot-Noise vs. Heisenberg Scaling

Classical estimation with n probes achieves standard deviation $\propto 1/\sqrt{n}$ (shot-noise limit). With n entangled quantum probes, the Quantum CRB allows $\propto 1/n$ (Heisenberg limit) — a quadratic improvement!

2 Quantum Cramér–Rao Bound

Q4 — What is the Quantum Fisher Information?

Question 4

Define the Quantum Fisher Information (QFI) $F_Q(\theta)$ and the Symmetric Logarithmic Derivative (SLD). Why is the SLD the natural quantum analogue of the classical score function?

Answer

In quantum mechanics, the state of the system is described by a density matrix $\rho(\theta)$ — a positive semi-definite operator with $\text{Tr}[\rho] = 1$. The parameter θ is encoded in the quantum state.

Symmetric Logarithmic Derivative (SLD): Because density matrices do not generally commute, we cannot simply define $\partial_\theta \ln \rho$. Instead, the SLD L_θ is defined *implicitly* by the symmetrized operator equation:

$$\frac{\partial \rho}{\partial \theta} = \frac{1}{2}(L_\theta \rho + \rho L_\theta). \quad (4)$$

Quantum Fisher Information:

$$F_Q(\theta) = \text{Tr}[\rho(\theta) L_\theta^2] \quad (5)$$

Why SLD is the right analogue: The classical score is $\partial_\theta \ln p$. Since $\partial_\theta p = p \cdot \partial_\theta \ln p$, the quantum analogue replaces $p \rightarrow \rho$ and $\partial_\theta \ln p \rightarrow L_\theta$, symmetrised to respect non-commutativity.

Spectral formula: In the eigenbasis $\rho = \sum_k \lambda_k |k\rangle\langle k|$:

$$F_Q(\theta) = 2 \sum_{j,k: \lambda_j + \lambda_k > 0} \frac{|\langle k | \partial_\theta \rho | j \rangle|^2}{\lambda_j + \lambda_k}.$$

Why “Symmetric” Logarithmic Derivative?

Quantum operators do not commute ($AB \neq BA$ in general), so $L_\theta \rho \neq \rho L_\theta$. The SLD uses the symmetrised product $\frac{1}{2}(L_\theta \rho + \rho L_\theta)$ to produce a Hermitian L_θ , ensuring all eigenvalues are real and the QFI is a well-defined non-negative number.

Q5 — State and derive the Quantum Cramér–Rao Bound.

Question 5

State the Quantum Cramér–Rao Bound (QCRB) and outline its derivation. In what sense is it a *stronger* result than the classical CRB?

Answer

The Quantum Cramér–Rao Bound:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{F_Q(\theta)} \quad (6)$$

Derivation sketch:

Step 1: Classical CRB for a fixed POVM $\{M_x\}$:

A quantum measurement $\{M_x\}$ (positive-operator valued measure) gives the probability $p(x | \theta) = \text{Tr}[\rho(\theta)M_x]$. By the classical CRB, $\text{Var}(\hat{\theta}) \geq 1/I(\theta; \{M_x\})$.

Step 2: Bound the classical Fisher information:

Using the Cauchy–Schwarz inequality for operators:

$$I(\theta; \{M_x\}) \leq F_Q(\theta).$$

Step 3: Optimise over all measurements:

$F_Q(\theta) = \sup_{\{M_x\}} I(\theta; \{M_x\})$, so:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta; \{M_x\})} \geq \frac{1}{F_Q(\theta)}.$$

Why it is stronger: The QCRB holds for *every* quantum measurement, not just a specific one. $F_Q(\theta)$ is therefore the *ultimate* (physics-limited) precision limit, optimised over all conceivable measurement strategies.

Optimal measurement: The bound is saturated by projective measurement in the eigenbasis of the SLD operator L_θ .

Q6 — QFI for pure states and role of entanglement.

Question 6

Write down the QFI for a pure state $|\psi(\theta)\rangle$. How does entanglement allow quantum metrology to surpass the classical shot-noise limit?

Answer

QFI for pure states:

For $\rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$:

$$F_Q(\theta) = 4 \left(\langle \partial_\theta \psi | \partial_\theta \psi \rangle - |\langle \psi | \partial_\theta \psi \rangle|^2 \right). \quad (7)$$

This is **four times the quantum geometric variance** of $|\psi\rangle$ with respect to θ -translations (related to the Fubini–Study metric on the space of quantum states).

Role of entanglement: Consider estimating a phase θ imprinted by a unitary $U(\theta) = e^{-i\theta H}$ on n probes.

Strategy	QFI	Best Std Dev
n separable (classical) probes	$F_Q = n (\Delta H)^2$	$\propto 1/\sqrt{n}$
n entangled (GHZ/NOON) probes	$F_Q = n^2 (\Delta H)^2$	$\propto 1/n$

The **Heisenberg limit** ($\propto 1/n$) is achievable only with entanglement. Coherence and quantum correlations effectively amplify the “slope” $\partial_\theta \rho$, boosting F_Q quadratically.

Real-world applications:

- **LIGO gravitational-wave detectors:** squeezed light states push beyond shot-noise in interferometric phase estimation.
- **Atomic clocks:** entangled spin states achieve sub-shot-noise frequency uncertainty.
- **MRI / NV centres:** quantum sensing of magnetic fields.

3 Comparison and Synthesis

Q7 — Classical vs. Quantum CRB: a structured comparison.

Question 7

Construct a detailed comparison of the Classical and Quantum Cramér–Rao Bounds, covering their information measures, physical interpretations, and the nature of the precision limits they impose.

Answer

Feature	Classical CRB	Quantum CRB
Bound	$\text{Var}(\hat{\theta}) \geq 1/I(\theta)$	$\text{Var}(\hat{\theta}) \geq 1/F_Q(\theta)$
Information	Fisher Info $I(\theta)$	QFI $F_Q(\theta)$
Data model	Probability dist. $p(x \theta)$	Density matrix $\rho(\theta)$
Measurement	Fixed / given	Optimised over all POVMs
Score function	$\partial_\theta \ln p(x \theta)$	SLD operator L_θ
Limit type	Statistical	Physical (quantum laws)
n-probe scaling	$1/\sqrt{n}$ (shot-noise)	$1/n$ possible (Heisenberg)
Enhancement	None beyond data	Entanglement & coherence
Achievability	MLE for exp. family	SLD eigenbasis measurement

Conceptual hierarchy:

$$\underbrace{\text{Var}(\hat{\theta})}_{\text{any estimator}} \geq \underbrace{\frac{1}{I(\theta; \{M_x\})}}_{\text{classical CRB for fixed } \{M_x\}} \geq \underbrace{\frac{1}{F_Q(\theta)}}_{\text{quantum CRB (ultimate limit)}}$$

Key conceptual difference:

- **Classical CRB:** the limit arises purely from *statistical randomness* in the data given a fixed measurement.
- **Quantum CRB:** the limit arises from *quantum mechanical constraints* on information extraction — Heisenberg’s uncertainty principle and the no-cloning theorem place fundamental caps on measurement precision.

Q8 — Why can the QFI exceed the classical Fisher information?

Question 8

Explain conceptually and mathematically why the Quantum Fisher Information $F_Q(\theta)$ is always greater than or equal to the classical Fisher information $I(\theta; \{M_x\})$ for any fixed measurement $\{M_x\}$.

Answer

The key inequality is:

$$I(\theta; \{M_x\}) \leq F_Q(\theta) \quad \text{for any POVM } \{M_x\}. \quad (8)$$

Proof sketch (Cauchy–Schwarz route):

Starting from $p(x | \theta) = \text{Tr}[\rho M_x]$:

$$\partial_\theta p = \text{Tr}[(\partial_\theta \rho) M_x] = \text{Tr} \left[\frac{L_\theta \rho + \rho L_\theta}{2} M_x \right] = \text{Re Tr}[\rho L_\theta M_x].$$

Applying Cauchy–Schwarz for the Hilbert–Schmidt inner product:

$$(\partial_\theta \ln p)^2 p = \frac{(\partial_\theta p)^2}{p} \leq \text{Tr}[\rho L_\theta^2 M_x]$$

Summing over x :

$$I(\theta; \{M_x\}) = \sum_x \frac{(\partial_\theta p)^2}{p} \leq \sum_x \text{Tr}[\rho L_\theta^2 M_x] = \text{Tr}[\rho L_\theta^2] = F_Q(\theta).$$

Physical meaning: A particular measurement extracts only a *projection* of the full quantum information encoded in $\rho(\theta)$. The QFI captures *all* information in the quantum state, accessible only by the optimal (SLD-eigenbasis) measurement. Every other measurement “discards” some information, hence $I \leq F_Q$.

4 Final Summary & Formula Sheet

Master Formula Sheet

Classical Cramér–Rao Bound

$$I(\theta) = \mathbb{E}[(\partial_\theta \ln p(x | \theta))^2] \quad (\text{Fisher Information})$$

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)} \quad (\text{CRB})$$

$$I_n(\theta) = n I_1(\theta) \quad (\text{i.i.d. scaling})$$

Quantum Cramér–Rao Bound

$$\partial_\theta \rho = \frac{1}{2}(L_\theta \rho + \rho L_\theta) \quad (\text{SLD definition})$$

$$F_Q(\theta) = \text{Tr}[\rho L_\theta^2] \quad (\text{QFI})$$

$$\text{Var}(\hat{\theta}) \geq \frac{1}{F_Q(\theta)} \quad (\text{QCRB})$$

$$F_Q = 4(\langle \partial_\theta \psi | \partial_\theta \psi \rangle \langle \partial_\theta \psi | \partial_\theta \psi \rangle - | \langle \psi | \partial_\theta \psi | \psi \rangle |^2) \quad (\text{pure state QFI})$$

Hierarchy

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta; \{M_x\})} \geq \frac{1}{F_Q(\theta)}$$

Scaling limits

$$\text{Shot-noise (classical): } \Delta\theta \propto \frac{1}{\sqrt{n}} \quad \text{Heisenberg (quantum): } \Delta\theta \propto \frac{1}{n}$$

Big-Picture Takeaway

1. The **classical CRB** sets the best precision achievable with any unbiased estimator, given a fixed classical data model. The limitation is *statistical*.
2. The **quantum CRB** extends this to quantum states, optimising over all possible measurements. The limitation is *physical* (laws of quantum mechanics).
3. **Quantum resources** — entanglement and coherence — can increase the QFI quadratically in the number of probes, enabling the Heisenberg limit $\Delta\theta \propto 1/n$, beating the classical $1/\sqrt{n}$.
4. Both bounds share the same mathematical spirit: $\text{Var} \geq 1/\text{Information}$. Larger information $\Rightarrow$ tighter bound $\Rightarrow$ more precise estimation.